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ELECTRIC POWER STEERING GEAR WITH AN ANTI-ROTATE FEATURE

Abstract

A steering assembly includes a housing. An interior wall of a cylindrical portion of the housing defines a groove that extends in an axial direction and having inwardly tapered walls. A ball screw disposed in the housing defines a hole in a radial direction. An anti-rotational pin having a tapered end is disposed in the hole, the tapered end corresponding to the inwardly tapered walls. A spring disposed in the hole biases the anti-rotational pin so that the tapered end of the anti-rotational pin engages the groove and restricts rotation of the ball screw. A support bushing disposed between the ball screw and the housing is fixedly coupled to the ball screw and defines a through hole between the groove and the hole of the ball screw, the through hole receives the anti-rotational pin. The support bushing absorbs radial loads exerted on the ball screw and supports the anti-rotational pin.

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Background/Summary

FIELD

[0001] The present disclosure relates to a steering gear for a vehicle with an anti-rotate feature.

BACKGROUND

[0002] Commercial vehicles that use Electric Powered Steering (EPS). Electric Powered Steering is implemented using an electric power steering gear. The electric power steering gear utilizes a ball nut to provide a ball screw with linear translation. This linear translation motion of the ball screw turns the tires to steer the path of the vehicle. To keep the ball screw from turning with the ball nut, an anti-rotational feature engages the gear housing to rotationally fix the ball screw relative to the housing. Current anti-rotational methods include a male spline on the ball screw and a female spline fixed to the gear housing. One such example is a pin in a hole through the ball screw that engages in an axial groove in the gear housing, permitting it to slide along the axial groove while prohibiting rotation relative to the gear housing. However, conventional methods do not prevent undesirable free rotational movement (referred to as lash) due to manufacturing tolerances and normal wear due to use.

SUMMARY

[0003] Current anti-rotational features permit a small amount of play between a ball screw and housing of a steering mechanism, also known as lash, due to normal wear and tear and manufacturing tolerances.

[0004] Embodiments of the present disclosure provide, in a first aspect, an electric powered steering assembly for a commercial vehicle, comprising: a housing including a cylindrical portion extending in an axial direction, an interior wall of the cylindrical portion defining a groove and extending in the axial direction, the groove having at least two inwardly tapered walls; a ball screw disposed in the housing, extending in the axial direction, and defining a hole extending in a radial direction; a ball nut disposed in the housing surrounding the ball screw and configured to rotate relative to the housing; an anti-rotational pin having a tapered end disposed in the hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls; a spring disposed in the hole and configured to bias the anti-rotational pin in a radially outward direction towards the groove so that the at least one tapered end of the anti-rotational pin engages the groove and thereby restricts rotation of the ball screw relative to the housing; and a support bushing disposed in the housing between the ball screw and the housing, the support bushing being fixedly coupled to the ball screw and defining a through hole disposed between the groove and the hole of the ball screw, the through hole configured to receive the anti-rotational pin, wherein the support bushing absorbs radial loads exerted on the ball screw by the electric power steering assembly and supports the anti-rotational pin.

[0005] According to an implementation of the first aspect, the interior wall of the cylindrical portion defining a second groove and extending in the axial direction, the second groove having at least two inwardly tapered walls; the ball screw disposed in the housing, extending in the axial direction, and defining a second hole extending in the radial direction; a second anti-rotational pin having a tapered end disposed in the second hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls of the second groove; and a second spring disposed in the second hole and configured to bias the second anti-rotational pin in a radially outward direction towards the second groove so that the at least one tapered end of the second anti-rotational pin engages the second groove and thereby restricts rotation of the ball screw relative to the housing.

[0006] According to an implementation of the first aspect, the at least two tapered walls of the groove inwardly taper at a first angle.

[0007] According to an implementation of the first aspect, the at least two tapered surfaces of the tapered end of the anti-rotational pin taper at the first angle to match the taper of the groove.
[0008] According to an implementation of the first aspect, the ball nut is configured to rotate over the ball screw and engage threads of the ball screw.
[0009] According to an implementation of the first aspect, the rotation of the ball nut over the ball screw results in an axial translation of the ball screw in a right or left direction based on a rotational direction of the ball nut.
[0010] According to an implementation of the first aspect, engaging the groove with the anti-rotational pin restricts the rotation of the ball screw relative to the ball nut.
[0011] According to an implementation of the first aspect, the support bushing is fixedly coupled to the ball screw using a plurality of bolts distributed evenly across the support bushing.
[0012] According to an implementation of the first aspect, the support bushing is composed of a smooth steel or bronze.
[0013] According to an implementation of the first aspect, a coating is applied on the support bushing.
[0014] According to an implementation of the first aspect, the coating is composed of polytetrafluoroethylene (PTFE).
[0015] A second aspect of the present disclosure provides a method of providing a lash-free electric powered steering gear for a commercial vehicle, the method comprising: providing a housing including a cylindrical portion extending in an axial direction, an interior wall of the cylindrical portion defining a groove and extending in the axial direction, the groove having at least two inwardly tapered walls; providing a ball screw disposed in the housing, extending in the axial direction, and defining a hole extending in a radial direction; providing a ball nut disposed in the housing surrounding the ball screw and configured to rotate relative to the housing; providing an anti-rotational pin having a tapered end disposed in the hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls; biasing the anti-rotational pin, using a spring disposed in the hole, in a radially outward direction towards the groove so that the at least one tapered end of the anti-rotational pin engages the groove and thereby restricts rotation of the ball screw relative to the housing; providing a support bushing disposed in the housing between the ball screw and the housing, the support bushing being fixedly coupled to the ball screw and defining a through hole disposed between the groove and the hole of the ball screw, wherein the through hole is configured to receive the anti-rotational pin; and absorbing, using the support bushing, radial loads exerted on the ball screw by the electric power steering assembly and supporting the anti-rotational pin.
[0016] According to an implementation of the second aspect, the method further comprises providing the interior wall of the cylindrical portion defining a second groove and extending in the axial direction, the second groove having at least two inwardly tapered walls; providing the ball screw disposed in the housing, extending in the axial direction, and defining a second hole extending in the radial direction; providing a second anti-rotational pin having a tapered end disposed in the second hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls of the second groove; and providing a second spring disposed in the second hole and configured to bias the second anti-rotational pin in a radially outward direction towards the second groove so that the at least one tapered end of the second anti-rotational pin engages the second groove and thereby restricts rotation of the ball screw relative to the housing.
[0017] According to an implementation of the second aspect, the at least two tapered walls of the groove inwardly taper at a first angle, and wherein the at least two tapered surfaces of the tapered end of the anti-rotational pin taper at the first angle to match the taper of the groove.
[0018] According to an implementation of the second aspect, the ball nut is configured to rotate over the ball screw and engage threads of the ball screw.

[0019] According to an implementation of the second aspect, the rotation of the ball nut over the ball screw results in an axial translation of the ball screw in a right or left direction based on a rotational direction of the ball nut.
[0020] According to an implementation of the second aspect, engaging the groove with the anti-rotational pin restricts the rotation of the ball screw relative to the ball nut.
[0021] According to an implementation of the second aspect, the support bushing is fixedly coupled to the ball screw using a plurality of bolts distributed evenly across the support bushing.
[0022] According to an implementation of the second aspect, the support bushing is composed of a smooth steel or bronze.
[0023] According to an implementation of the second aspect, a coating is applied on the support bushing, and wherein the coating is composed of polytetrafluoroethylene (PTFE).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] Embodiments of the present disclosure will be described in even greater detail below based on the exemplary figures. The present disclosure is not limited to the exemplary embodiments. All features described and/or illustrated herein can be used alone or combined in different combinations in embodiments of the present disclosure. The features and advantages of various embodiments of the present disclosure will become apparent by reading the following detailed description with reference to the attached drawings which illustrate the following:
[0025] FIG. **1** illustrates an overview of a gear assembly of a vehicle, according to one or more examples of the present disclosure;
[0026] FIG. **2** illustrates a side cross-section view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0027] FIG. **3** illustrates a front cross-section view of a housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0028] FIG. **4** illustrates another front cross-section view of the housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0029] FIG. **5** illustrates a perspective view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0030] FIG. **6** illustrates a perspective view of a cross-section of a housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0031] FIG. **7** illustrates a view of an anti-rotational pin, according to one or more examples of the present disclosure;
[0032] FIG. **8**A illustrates another view of the anti-rotational pin, according to one or more examples of the present disclosure;
[0033] FIG. **8**B illustrates another view of the anti-rotational pin, according to one or more examples of the present disclosure;
[0034] FIG. **8**C illustrates a perspective view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0035] FIG. **9** illustrates another overview of the gear assembly of a vehicle, according to one or more examples of the present disclosure;
[0036] FIG. **10** illustrates another side cross-section view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0037] FIG. **11**A illustrates another cross-section view of the housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure;
[0038] FIG. **11**B illustrates a partial cross-section view of the housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure; and

[0039] FIG. **12** illustrates a support bushing associated with the gear assembly of the vehicle, according to one or more examples of the present disclosure.

DETAILED DESCRIPTION

[0040] Examples of the presented application will now be described more fully hereinafter with reference to the accompanying FIGs., in which some, but not all, examples of the application are shown. Indeed, the application may be exemplified in different forms and should not be construed as limited to the examples set forth herein; rather, these examples are provided so that the application will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term “a” and/or “an” shall mean “one or more” even though the phrase “one or more” is also used herein. Furthermore, when it is said herein that something is “based on” something else, it may be based on one or more other things as well. In other words, unless expressly indicated otherwise, as used herein “based on” means “based at least in part on” or “based at least partially on”.

[0041] Commercial vehicles use Electric Powered Steering (EPS) gears for steering. EPS gears include a ball screw oriented transversely to the vehicle and disposed within a gear housing and a ball nut surrounding the ball screw in the housing. Upon receiving steering commands from a driver, the EPS gear of the vehicle comprising the ball nut and ball screw is configured transfer the steering command from the driver to the wheels of the vehicle. For example, upon receiving the steering command from a driver, the ball nut is configured to rotate around the ball screw. Due to the threading of the ball screw and ball nut, rotation of the ball nut causes the ball screw to translate in the right or left direction depending on the rotational direction of rotation of the ball nut. The translation of the ball screw causes the wheels of the vehicle to turn, thereby steering the vehicle. In order to facilitate the translation of the ball screw, the ball screw has an anti-rotational feature that prevents the ball screw from rotating relative to the housing. However, due to manufacturing tolerances and normal wear, conventional anti-rotational feature permits a small amount of play between the ball screw and housing, also known as lash.

[0042] One possible way to avoid lash in the electric power steering gear is to use anti-rotational pins having tapered ends with the ball screw. In some embodiments, two anti-rotational pins with tapered ends are placed in a common through hole that extends radially through the ball screw so that their tapered ends face in a radially outward direction from the center of the ball screw. The inner cylindrical wall of the housing includes two grooves extending axially that are configured to receive the tapered ends of the two anti-rotational pins. The grooves include tapered groove walls corresponding to the tapered ends of the anti-rotational pins. The angle of tapering in the grooves is same as the angle of tapering on the two anti-rotational pins. A spring is placed between the two anti-rotational pins in the ball screw so as to bias the two anti-rotational pins in the radially outward direction and to hold the anti-rotational pins in contact with the grooves and to ensure a tight fit between pins and the grooves. This arrangement of the anti-rotational pins prevents the ball screw from rotating relative to the housing and eliminates lash, while permitting axial movement of the ball screw relative to the housing. This embodiment is discussed in more detail with respect to FIGS. **1-5**.

[0043] In another embodiment, any number of anti-rotational pins may be used to avoid lash in the electric powered steering gears. The inner cylindrical wall of the housing may be modified to include the same number of grooves extending in the axial direction, as the number of anti-rotational pins used. In this embodiment, instead of a through hole, the ball screw has a hole for each anti-rotational pin and a respective spring to bias the anti-rotational pin in a radially outward direction towards a corresponding groove. This embodiment is discussed in more detail in FIGS. **6-12**.

[0044] FIG. **1** illustrates an overview of a gear assembly of a vehicle, according to one or more examples of the present disclosure. The gear assembly **100** depicted in FIG. **1** is responsible for providing the steering functionality to the vehicle. Section **102** of the gear assembly **100** shown in FIG. **1** comprises a ball screw and a ball nut, which is discussed in more detail in FIGS. **2-5**.

[0045] FIG. **2** illustrates a side cross-section view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure. Portion **102** of the gear assembly **100** includes a housing **104** of the gear assembly that is cylindrically shaped and extends transversely to the vehicle. A ball screw and ball nut (shown in more detail in FIGS. **3** and **5**) are present in the housing **104**. The ball nut and the ball screw are configured to convey the steering commands received from a driver to the wheels of the vehicle to steer the vehicle. In order to hold the ball screw grounded to the housing **104**, the housing **104** includes two tapered grooves to receive corresponding anti-rotational pins. The grooves of the housing **104** are tapered at the same angle as the taper of the anti-rotational pins **108** and **106** and extend the length of the housing **104**. Anti-rotational pins **106** and **108** are disposed in a through-hole that extends in a radial direction through the ball screw. A spring **110** is disposed between the two anti-rotational pins **106** and **108** to bias the anti-rotational pins **106** and **108** in a radially outward direction so that the pins engage with the corresponding grooves in the housing **104**. By engaging with the grooves, the anti-rotational pins **106** and **108** prevent the ball screw from rotating relative to the housing when the ball nut rotates, but permit the ball screw to translate axially in the housing. When the ball nut rotates and the ball screw is held stationary using the anti-rotational pins **106** and **108**, the relative rotation between the ball nut and the ball screw causes translation of the ball screw in an axial direction relative to the housing. Because the tapered ends of each anti-rotational pin is biased toward the groove having corresponding tapered sides, a constant tight fit is ensured, thereby eliminating lash of the ball screw relative to the housing.

[0046] FIG. **3** illustrates a front cross-section view of a housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure. As disclosed with respect to FIG. **2**, the housing **104** includes a ball screw **310** and the ball nut **312**. The housing **104** of the gear assembly includes two tapered grooves on opposite ends. The angle of taper of the grooves in the housing **104** of the gear assembly matches the angle of taper on the anti-rotational pins **106** and **108**. Two anti-rotational pins **106** and **108** are placed in a through hole of the ball screw **310** and inserted in the two tapered grooves of the housing **104**. A spring **110** is placed between the two anti-rotational pins **106** and **108** to provide a radial outward force to hold the anti-rotational pins **106** and **108** in constant contact with the housing **104** of the gear assembly. The forces applied on the anti-rotational pins **106** and **108**, the ball screw **310**, and the ball nut **312** are discussed in more detail in FIG. **4**.

[0047] FIG. **4** illustrates another front cross-section view of a housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure. FIG. **4** is similar to FIG. **3**, except that FIG. **4** discusses the forces at play between the anti-rotational pins **106** and **108**, the ball screw **310**, the ball nut **312**, and the housing **104** of the gear assembly, as shown in FIG. **3**. Arrows **402** highlight the taper angle of the anti-rotational pins **106** and **108** and the corresponding grooves in the housing **104**. The taper angle of the anti-rotational pins **106** and **108** is similar to the taper angle of the corresponding grooves in the housing **104**. The spring **110** placed between the anti-rotational pins **106** and **108** provides a radial force highlighted by arrow **404** that ensures that the anti-rotational pins **106** and **108** are constantly in contact with the grooves created in the housing **104**. Because the end of the groove is slightly narrower than the end of anti-rotational pins, spring ensures a constant tight fit between the tapered surfaces of each pin and the tapered walls of each groove. Arrow **406** depicts the rotational direction of the ball nut **312**, which may be in either direction. As the ball nut **312** rotates relative to the ball screw, the threading of the ball nut and the ball screw causes, the ball screw to translate in an axial direction.

[0048] FIG. **5** illustrates a perspective view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure. Perspective view **500** depicts a housing **104** of the gear assembly. The housing **104** of the gear assembly is cylindrically shaped and extends in an axial direction. Ball screw **310** extends axially within the housing **104**. Ball nut **312** is configured to rotate relative to the ball screw **310** and engage the threads of the ball screw **310**. In some embodiments, the ball nut **312** may be housed in a portion of the housing **104** that permits rotation of the ball nut **312**, while restricting translation of the ball nut **312**. Anti-rotational pins **106** and **108** are placed in a through hole that extends radially through the ball screw **310**. Spring **110** placed between the anti-rotational pins **106** and **108** provides an outward radial force that ensures that the anti-rotational pins **106** and **108** are in constant tight contact with the grooves **502** and **504** in an inner wall of the housing **104** that extend axially. Grooves **502** and **504** are created on opposite sides of the ball screw **310** corresponding to the open ends of the through-hole in the ball screw and have tapered side walls that taper at the same angle as the tapered surfaces of the anti-rotational pins **106** and **108**. Grooves **502** and **504** extend axially along the length of the cylindrical portion of the housing **104**. The anti-rotational pins **106** and **108** engaging with the grooves **502** and **504** of the housing **104** restrict any rotation of the ball screw **310** relative to the housing. Thus, because the rotation of the ball screw **310** is restricted, the circumferential rotation of the ball nut **312** results in translation of the ball screw in the axial direction.

[0049] In some embodiments, any number of anti-rotational pins may be used to avoid lash in the electric powered steering gears. The inner cylindrical wall of the housing may be modified to include the same number of grooves extending in the axial direction, as the number of anti-rotational pins used. In this embodiment, instead of a through hole, the ball screw has a hole for each anti-rotational pin and a respective spring to bias the anti-rotational pin in a radially outward direction towards a corresponding groove.

[0050] FIG. **6** illustrates another cross-section view of a housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure. As disclosed with respect to FIG. **3**, the housing **104** includes a ball screw **310** and the ball nut **312**. The cross-section view **600** of FIG. **6** is similar to cross-section view **300** of FIG. **3** except the cross-section view **600** has four tapered grooves **602**, **604**, **606**, and **608** distributed in the housing **104** instead of the two shown in FIG. **3**. In some embodiments, the number of tapered grooves in the housing **104** may be selected based on the number of anti-rotational pins used to prevent lash in the electric power steering gears. Instead of a through hole, the ball screw may include four separate holes, where each hole of the ball screw **310** corresponds to a corresponding tapered groove **602**, **604**, **606**, or **608** in the housing **104**. Anti-rotational pins **106**, **108**, **610**, and **612** are placed in the holes of the ball screw **310** and a spring is placed in each hole, at a first end of the anti-rotational pin, to bias the tapered ends of each anti-rotational pin in a radially outward direction to engage with the four tapered grooves **602**, **604**, **606**, and **608** of the housing **104**. For example, FIG. **6** shows springs **110**, **614**, **616**, and **618** that are inserted at the end of anti-rotational pins **106**, **108**, **610** and **612** in the ball screw **310**. As discussed previously the springs **110**, **614**, **616**, **618** provide a radial outward force to hold the anti-rotational pins **106**, **108**, **610**, and **612** in constant contact with the housing **104** of the gear assembly.

[0051] A durability and performance of the anti-rotational pins **106**, **108**, **610**, and **612** may be extended by modifying the anti-rotational pin by applying a coating or affixing a bearing material on the tapered ends of the anti-rotational pins. The details of the coating applied on the anti-rotational pins and the bearing material affixed on the anti-rotational pins are discussed in more detail in FIGS. **7** and **8**A-C respectively.

[0052] FIG. **7** illustrates a view of an anti-rotational pins according to one or more examples of the present disclosure. Illustration **700** shows a view of an anti-rotational pin **106** in which a low friction coating **702** is applied on the tapered ends of the anti-rotational pin **106** that comes in contact with the tapered grooves of the housing **104**. In some embodiments, the low friction coating enhances the life of the anti-rotational pin **106** by reducing the friction between the tapered ends of the anti-rotational pin **106** and the tapered groove of the housing **104**. For example, a coefficient of friction between the low friction coating **702** applied on the tapered ends of the anti-rotational pin

106 and the tapered walls of the groove of the housing **104** is less than a coefficient of friction between the tapered ends of the anti-rotational pin **106** and the tapered walls of the grooves of the housing **104**. The low friction coating **702** applied on the tapered ends of the anti-rotational pin **702** may have a thickness in the range of 0.0001 inches to 0.0005 inches. In some other examples, the thickness of the coating may be approximately 50 microns. In some cases, the low friction coating **702** may comprise polytetrafluoroethylene (PTFE). Additionally and/or alternatively, the low friction coating may be Turcite (a combination of acetal and TFE), polyeretherketone (PEEK), ultra-high molecular weight polyethylene, non-ferrous based material like aluminized bronze, or polyphenylene sulphide (PPS). The low friction coating **702** may be applied to the anti-rotational pin **106** by spraying, electro-plating, hot dipping (galvanizing), vacuum deposition, and/or baking. In some embodiments, a low friction coating of the same material may also be applied on the tapered groove of the housing **104** to reduce friction between the tapered ends of the anti-rotational pin **106** and the tapered walls of the groove, and extend the life of the anti-rotational pin **106**.

[0053] FIG. **8**A illustrates another view of an anti-rotational pins according to one or more examples of the present disclosure. Illustration **800** shows a view of an anti-rotational pin **106** in which the tapered ends of the anti-rotational pin **106** are covered with a bearing material **802**. In some embodiments, the bearing material enhances the life of the anti-rotational pin **106** by reducing the friction between the tapered ends of the anti-rotational pin **106** and the tapered groove of the housing **104**. For example, a coefficient of friction between the bearing surface **802** applied on the tapered ends of the anti-rotational pin **106** and the tapered walls of the groove of the housing **104** is less than a coefficient of friction between the tapered ends of the anti-rotational pin **106** and the tapered walls of the grooves of the housing **104**. In some embodiments, the coating **802** may be affixed to the anti-rotational pin **106** using a screw **804**. As shown in FIG. **8**A, the screw **804** may be attached to the anti-rotational pin **106** on a side of the anti-rotational pin **804** that is between the tapered ends and adjacent to a smaller side of the tapered ends. In some cases, a second screw (not shown) may be used on an opposite side to securely affix the bearing material to the tapered ends of the anti-rotational pin **106**. In some embodiments, the bearing material may be affixed to the tapered ends of the anti-rotational pin **106** by means of an adhesive. In some embodiments, the bearing material may be held in a recessed “pocket” machined in faces of the tapered end of the anti-rotational pin. In such embodiments, the depth of the pocket is designed to “trap” the bearing material. In some embodiments, two “L-shaped” pieces of bearing material **802** may be affixed to cover each of the tapered surfaces of the tapered end of the anti-rotational pin **106**.

[0054] Additionally and/or alternatively, bearing material may also be applied on the tapered groove of the housing **104** to reduce friction between the tapered ends of the anti-rotational pin **106** and the tapered walls of the groove, and extend the life of the anti-rotational pin **106**. In some embodiments, the bearing material may be held in a recessed “pocket” machined in faces of the tapered grooves of the housing **104**. The depth of the pocket is designed to “trap” the bearing material.

[0055] FIG. **8**B illustrates another view of the anti-rotational pin, according to one or more examples of the present disclosure. The view **850** of anti-rotational pin **106** shown in FIG. **8**B depicts an alternative way of affixing the bearing material on the surface of the tapered ends of the anti-rotational pin **106**. As shown in view **850** of FIG. **8**B, a recessed pocket may be created on the surface **852** of the tapered ends to reveal a second tapered surface **854**. The bearing material **802** is configured to abut the second tapered surface **854** and fit in the recessed pocket of the tapered surface **852** of the anti-rotational pin **106**. In order for the bearing material **802** to be securely held in the anti-rotational pin **106**, the bearing material **802** may have dimensions same as the dimensions of the recessed pocket. The bearing material **802** may also have a thickness that is same as the thickness of the recessed pocket. In some embodiments, the bearing material **802** is held in place by being trapped between the recessed pocket and the corresponding tapered groove in the housing **104**.

[0056] FIG. **8**C illustrates a perspective view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure. Perspective view **875** depicts the housing **104** of the gear assembly that is cylindrically shaped and extends in an axial direction. Ball screw **310** extends axially within the housing **104**. Anti-rotational pins **106**, **610**, and **612** are placed in holes in the ball screw **310**. Springs **110**, **616**, and **618** (not shown) placed between the anti-rotational pins **106**, **610**, and **612** and the holes that provides an outward radial force that ensures that the anti-rotational pins **106**, **610**, and **612** are in constant tight contact with the respective grooves in an inner wall of the housing **104** that extend axially. As shown in the perspective view **875**, a bearing material **802** is affixed on the tapered surface of the anti-rotational pins **106**, **610**, and **612**.

[0057] The bearing material may be a steel backed sheet with bronze and polymer fillers. In some embodiments, the bearing material may be provided as a sheet of steel with bronze layers. The bronze layers may have holes in them, which are filled with filler material such as graphite, or PTFE. These layers are then rolled up into a bushing and used as the material to create a bearing surface. Additionally and/or alternatively, the bearing material **802** may comprise polytetrafluoroethylene (PTFE) or other suitable materials.

[0058] In some embodiments, a support bushing may be used to support and stabilize the ball screw. The support bushing is an approvement that may absorb external radial loads applied on the ball screw and in doing so, reduce the radial movement of the ball screw and the anti-rotational pins which reduces the stress on the anti-rotational pins.

[0059] FIG. **9** illustrates another overview of a gear assembly of a vehicle, according to one or more examples of the present disclosure. The gear assembly **900** of FIG. **9** is similar to the gear assembly **100** shown in FIG. **1**. As discussed with respect to FIG. **1**, the gear assembly **900** depicted in FIG. **1** is responsible for providing the steering functionality to the vehicle. When a steering command is received at the gear assembly **900**, the steering command is transmitted via tie rods **902** of the gear assembly **900** to the wheels connected to the gear assembly **900**. In some embodiments, the steering of the wheels of the vehicle exerts an external load on the ball-screw **310** via the tie rods **902**. For example, the movement of the wheels in response to the steering command induces an arc-like motion in the tie rods **902**. The arc-like motion of the tie-rods **902** may change an angle of connection between the tie-rods **902** and a ball joint that connects the tie-rods to the ball screw **310**. The change in the angle of connection of the ball joint may exert a load on the ball screw **310**. Additionally, a suspension of the vehicle, also attached to the wheels of the vehicle may exert an external load on the ball screw **310**. The external loads, stemming from the tie-rods **902** or the suspension of the vehicle, may manifest as radial loads on the ball screw **310**. This is discussed in more detail in FIGS. **10-12**.

[0060] FIG. **10** illustrates another side cross-section view of a portion of the complete gear assembly of the vehicle, according to one or more examples of the present disclosure. Portion **1000** of the gear assembly **900** includes a housing **104** of the gear assembly that is cylindrically shaped and extends transversely to the vehicle. A ball screw and ball nut (shown in more detail in FIGS. **3** and **5**) are present in the housing **104**. The ball nut and the ball screw are configured to convey the steering commands received from a driver to the wheels of the vehicle to steer the vehicle. In order to hold the ball screw grounded to the housing **104**, the housing **104** includes tapered grooves to receive corresponding anti-rotational pins **108** and **106**. The grooves of the housing **104** are tapered at the same angle as the taper of the anti-rotational pins **108** and **106** and extend the length of the housing **104**. Anti-rotational pins **106** and **108** are disposed in independent holes that extend in a radial direction. Spring **614** and **110** is disposed between the anti-rotational pin **106** and **108** and the surface of the corresponding hole to bias the anti-rotational pin **106** and **108** in a radially outward direction so that the pins engage with the corresponding grooves in the housing **104**. By engaging with the grooves, the anti-rotational pins **106** and **108** prevent the ball screw from rotating relative to the housing when the ball nut rotates, but permit the ball screw to translate axially in the

housing. When the ball nut rotates and the ball screw is held stationary using the anti-rotational pins **106** and **108**, the relative rotation between the ball nut and the ball screw causes translation of the ball screw in an axial direction relative to the housing. Because the tapered ends of each anti-rotational pin are biased toward the groove having corresponding tapered sides, a constant tight fit is ensured, thereby eliminating lash of the ball screw relative to the housing. In some embodiments, a radial load is exerted on the ball screw **310** based on the arc-like motion of the tie-rods **902** that causes changes an angle of connection between the tie-rods **902** and a ball joint that connects the tie-rods to the ball screw **310**. Additionally, radial loads may be exerted on the ball screw **310** by the suspension of the vehicle that is connected to the gear assembly (not shown).
[0061] In the absence of a support bushing, the external loads that are exerted on the ball screw may be exerted on the anti-rotational pin **106** and **108**. The external loads exerted on the ball screw may cause radial motion of the ball screw, which in turn causes radial movement of the anti-rotational pins **106** and **108** which may increase wear and tear of the anti-rotational pins **106** and **108** and thereby reduce the efficiency of the anti-rotational pins **106** and **108** in preventing lash during steering.
[0062] A support bushing **1004** that is installed around the anti-rotational pins may absorb the radial loads applied on the ball screw **310**. By absorbing the radial loads, the support bushing **1004** reduces the radial movement of the ball screw **310** and stabilizes the ball screw. By stabilizing the ball screw, the support bushing **1004** is able to reduce the radial movements of the anti-rotational pins **106** and **108**. Once the radial movement of the anti-rotational pins **106** and **108** is reduced, the anti-rotational pins **106** and **108** may only be responsive to the torque exerted on the ball screw **310** in response to the steering command received from the electric powered steering gears, thereby working efficiently to hold the ball screw **310** with respect to the ball nut **312**.
[0063] FIG. **11**A illustrates another cross-section view of the housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure. View **1100** of FIG. **11**A shows the position of the support bushing **1004** in the housing **104** of the gear assembly **900** (shown in FIG. **9**). View **1100** depicts a housing **104** which houses the ball nut **312** (not shown in FIG. **11**A), the ball screw **312** (not shown in FIG. **11**A), anti-rotational pins **106**, **108**, **610**, and **612** placed in tapered grooves of the housing **104**, and springs **110**, **614**, **616**, and **618** (**616** and **618** are not shown in FIG. **11**A) corresponding each of the anti-rotational pins **108**, **106**, **610**, and **612** (**610** and **612** not shown in FIG. **11**A) respectively. The support bushing **1004** is disposed between the housing **104** and the ball screw **310**. In some embodiments, the support bushing **1004** may include holes corresponding to the tapered grooves of the housing **104** and holes in the ball screw **310**. As shown in view **1100** of FIG. **11**A, the holes of the support bushing **1004** receive the anti-rotational pins **106**, **108**, **610**, and **612**.
[0064] FIG. **11**B illustrates a partial cross-section view of the housing of the gear assembly of the vehicle, according to one or more examples of the present disclosure. View **1150** of FIG. **11**B provides a three-dimensional (3D) view of the support bushing **1004** without the housing **104**. From view **1150**, it is observed that the support bushing **1004** is affixed to the ball screw **310** using a number of bolts **1102**. View **1150** of FIG. **11**B also shows the tapered ends of anti-rotational pins **106** and **610** that are put through the holes of the support bushing **1004** into the ball screw **310**. The tapered ends of the anti-rotational pins **106** and **610** are biased against the tapered walls of the respective grooves of the housing **104** (not shown in view **1150**).
[0065] FIG. **12** illustrates a support bushing associated with the gear assembly of the vehicle, according to one or more examples of the present disclosure. Support bushing **1004** shown in FIG. **12** is circular in shape and is designed to surround the ball screw **310**. The support bushing **1004** may include differently-shaped holes **1202** and **1204**. The holes **1202** of the support bushing **1004** are configured to receive the anti-rotational pins. In some embodiments, the number of holes **1202** provided in the support bushing **1004** may be dependent on a number of anti-rotational pins installed on the ball screw **310**. In some embodiments, the holes **1202** have a sliding contact to

accommodate the tapered ends of the anti-rotational pins **106**, **610**, **612**, and **614** (not shown in FIG. **12**). The holes **1204** of the support bushing **1004** are configured to receive bolts **1102** that attach the support bushing **1004** to the ball screw **310**. As with respect to the number of holes **1202**, the number of holes **1204** may be provided in the support bushing **1004** may be dependent on a number of bolts **1102** that are used to fasten the support bushing **1004** to the ball screw **310**. Additionally and/or alternatively, the support bushing **1004** may be affixed to the ball screw **310** using pins, retaining rings or machined shoulder, or other mechanisms.

[0066] The support bushing **1004** may be composed of a smooth steel, a bearing metal (e.g., bronze), or a bearing material as described with respect to FIG. **8**A. In some embodiments, a coating may be applied on the support bushing **1004**. The coating may be applied as discussed in detail with respect to FIG. **7**. The application of the coating may enhance the durability of the support bushing by reducing a coefficient of friction between the support bushing **1004** and the ball screw **310**.

[0067] While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

[0068] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. An electric powered steering assembly for a commercial vehicle, comprising: a housing including a cylindrical portion extending in an axial direction, an interior wall of the cylindrical portion defining a groove and extending in the axial direction, the groove having at least two inwardly tapered walls; a ball screw disposed in the housing, extending in the axial direction, and defining a hole extending in a radial direction; a ball nut disposed in the housing surrounding the ball screw and configured to rotate relative to the housing; an anti-rotational pin having a tapered end disposed in the hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls; a spring disposed in the hole and configured to bias the anti-rotational pin in a radially outward direction towards the groove so that the at least one tapered end of the anti-rotational pin engages the groove and thereby restricts rotation of the ball screw relative to the housing; and a support bushing disposed in the housing between the ball screw and the housing, the support bushing being fixedly coupled to the ball screw and defining a through hole disposed between the groove and the hole of the ball screw, the through hole configured to receive the anti-rotational pin, wherein the support bushing absorbs radial loads exerted on the ball screw by the electric power steering assembly and supports the anti-rotational pin.

2. The electric powered steering assembly of claim 1, further comprising: the interior wall of the cylindrical portion defining a second groove and extending in the axial direction, the second groove having at least two inwardly tapered walls; the ball screw disposed in the housing, extending in the axial direction, and defining a second hole extending in the radial direction; a second anti-rotational pin having a tapered end disposed in the second hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls of the second groove; and a second spring disposed in the second hole and configured to bias the second anti-rotational pin in a radially outward direction towards the second groove so that the at least one tapered end of the second anti-rotational pin engages the second groove and thereby restricts rotation of the ball screw relative to the housing.

3. The electric powered steering gear of claim 1, wherein the at least two tapered walls of the groove inwardly taper at a first angle.

4. The electric powered steering gear of claim 3, wherein the at least two tapered surfaces of the tapered end of the anti-rotational pin taper at the first angle to match the taper of the groove.

5. The electric powered steering gear of claim 1, wherein the ball nut is configured to rotate over the ball screw and engage threads of the ball screw.

6. The electric powered steering gear of claim 5, wherein the rotation of the ball nut over the ball screw results in an axial translation of the ball screw in a right or left direction based on a rotational direction of the ball nut.

7. The electric powered steering gear of claim 1, wherein engaging the groove with the anti-rotational pin restricts the rotation of the ball screw relative to the ball nut.

8. The electric powered steering gear of claim 1, wherein the support bushing is fixedly coupled to the ball screw using a plurality of bolts distributed evenly across the support bushing.

9. The electric powered steering gear of claim 1, wherein the support bushing is composed of a smooth steel or bronze.

10. The electric powered steering gear of claim 1, further comprising applying a coating on the support bushing.

11. The electric powered steering gear of claim 1, wherein the coating is composed of polytetrafluoroethylene (PTFE).

12. A method of providing a lash-free electric powered steering gear for a commercial vehicle, the method comprising: providing a housing including a cylindrical portion extending in an axial direction, an interior wall of the cylindrical portion defining a groove and extending in the axial direction, the groove having at least two inwardly tapered walls; providing a ball screw disposed in the housing, extending in the axial direction, and defining a hole extending in a radial direction; providing a ball nut disposed in the housing surrounding the ball screw and configured to rotate relative to the housing; providing an anti-rotational pin having a tapered end disposed in the hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls; biasing the anti-rotational pin, using a spring disposed in the hole, in a radially outward direction towards the groove so that the at least one tapered end of the anti-rotational pin engages the groove and thereby restricts rotation of the ball screw relative to the housing; providing a support bushing disposed in the housing between the ball screw and the housing, the support bushing being fixedly coupled to the ball screw and defining a through hole disposed between the groove and the hole of the ball screw, wherein the through hole is configured to receive the anti-rotational pin; and absorbing, using the support bushing, radial loads exerted on the ball screw by the electric power steering assembly and supporting the anti-rotational pin.

13. The method of claim 12, further comprising: providing the interior wall of the cylindrical portion defining a second groove and extending in the axial direction, the second groove having at least two inwardly tapered walls; providing the ball screw disposed in the housing, extending in the axial direction, and defining a second hole extending in the radial direction; providing a second anti-rotational pin having a tapered end disposed in the second hole, the tapered end having at least two tapered surfaces, each corresponding to a respective one of the at least two inwardly tapered walls of the second groove; and providing a second spring disposed in the second hole and configured to bias the second anti-rotational pin in a radially outward direction towards the second groove so that the at least one tapered end of the second anti-rotational pin engages the second groove and thereby restricts rotation of the ball screw relative to the housing.

14. The method of claim 12, wherein the at least two tapered walls of the groove inwardly taper at a first angle, and wherein the at least two tapered surfaces of the tapered end of the anti-rotational pin taper at the first angle to match the taper of the groove.

15. The method of claim 12, wherein the ball nut is configured to rotate over the ball screw and engage threads of the ball screw.

16. The method of claim 15, wherein the rotation of the ball nut over the ball screw results in an axial translation of the ball screw in a right or left direction based on a rotational direction of the ball nut.

17. The method of claim 1, wherein engaging the groove with the anti-rotational pin restricts the rotation of the ball screw relative to the ball nut.

18. The method of claim 1, wherein the support bushing is fixedly coupled to the ball screw using a plurality of bolts distributed evenly across the support bushing.

19. The method of claim 1, wherein the support bushing is composed of a smooth steel or bronze.

20. The method of claim 1, further comprising applying a coating on the support bushing, and wherein the coating is composed of polytetrafluoroethylene (PTFE).